Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Starlight
Project / Solution Name	FreshPlate
Name (Leader)	WANG HAOTONG
Email & Phone (Leader)	EEE2509044@xmu.edu.my | +60176130343
University Affiliation(s)	Xiamen University Malaysia Team members: 1
Live Prototype URL	https://fresh-plate-psi.vercel.app
Code Repository URL (public / access granted to judges)	https://github.com/ShatteredDreams819726/FreshPlate

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

<table>
<tr><td colspan="3">Vercel serves FreshPlate's React interface and a Node.js extraction function. Selected stock or recipe text, or one JPG/PNG, is sent to Gemini. Returned fields enter editable preview and validation; only separate human confirmation saves stock or recipes.

Versioned browser localStorage holds ingredients, batches, recipes, plan snapshots and activity. A deterministic planner shares available stock across dishes, respects existing-order reservations and prep caps, and prioritizes earlier-expiring eligible batches.

Approval revalidates the saved allocation and reserves stock without deducting it. Completion deducts approved quantities and releases plan reservations; cancellation releases those reservations. Validated transitions update inventory and Activity Log together. Only one approved, unfinished plan is allowed.</td></tr>
<tr><td>Text / JPG / PNG →</td><td>Vercel Function ↔</td><td>Gemini API</td></tr>
<tr><td colspan="3">Editable preview → Validate → Confirm → Browser storage</td></tr>
<tr><td colspan="3">Shared planner → Approve / Complete / Cancel → Stock and log</td></tr>
</table>

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript, HTML and CSS	Interface, import validation, allocation and state logic.	Language standards; implementations have their own licenses.
React / React DOM 19.3.1	Application interface and rendering.	MIT.
Vite 8.3.5; React plugin 6.1.2	Development server and frontend build.	MIT and bundled dependency notices.
Node.js 24.22.1; npm 11.19.1 (local)	Server function, CLI planner, automated checks and package management.	Node: MIT and bundled notices. npm: Artistic-2.0 and dependency terms.

Name & Version	Purpose in Project	License
Supporting runtime/build libraries	Scheduler; Rolldown, PostCSS and related build packages.	MIT; Lightning CSS MPL-2.0; detect-libc Apache-2.0; picocolors ISC; source-map-js BSD-3-Clause. Bundled notices apply.
Authoring tools (not app runtime)	VS Code; Word/PowerPoint 16.0; Codex artifact-tool 2.8.99; Python 3.12 / python-docx 1.2.0.	Microsoft/OpenAI proprietary terms; Python PSF; python-docx MIT.
Authoring support libraries	lxml, Pillow, pypdf, qrcode/zxing-cpp, colorama, pdf2image; Poppler and Playwright/Edge for QA.	Applicable MIT/BSD/Apache-2.0 licenses; Poppler GPL; Edge Microsoft terms.

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini API — Google	Text/image extraction of stock and recipes into JSON drafts through the server extraction endpoint.	Free tier.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosts the React/Vite interface and Node.js extraction function.	Basic plan
Replit	Runs the shared core planner for two simulation scenarios.	Replit Core
GitHub	Remote source repository.	Basic plan
ChatGPT / Codex — OpenAI	Development and material preparation; not called by the application.	ChatGPT Plus subscription;

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-3.5-flash-lite	Google Gemini API	Hosted pre-trained model for stock/recipe text and image extraction.	Google APIs Terms and Gemini API Additional Terms.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Built-in stock and recipes	Project-created examples	Synthetic batches, dates, reservations and recipes.	Project-authored.	N
Planning scenarios	Project-created evaluation fixtures	Synthetic default / zero tomato-egg cap; 61 / 45 servings.	Project-authored.	N
Import test examples	Project test materials	Synthetic text, mock responses and fictional stock/recipe images.	Project examples;	N
Product and pitch visuals	Prototype screenshots and ChatGPT image generation	Self-captured synthetic UI; generated food/restaurant illustrations.	Prototype screenshots; generated assets under OpenAI terms.	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT — Plus subscription	Problem framing, requirements, content, documents and pitch/speech preparation; image generation for visual references and selected illustrations.	Moderate
Codex — access through Plus	Code implementation and iteration, tests, debugging, import/planning workflows, documentation and PPT drafts.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

I defined the requirements and goals, then reviewed, edited and accepted the code, text, interfaces and visuals. I ran the application, checked text/image imports and other operations, and completed online functional acceptance. I verified outputs against source materials and finalized the documents and slides. Automated checks cover extraction failures, validation, allocation, reservations, state transitions and persistence.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Sponsor platform execution environment.	Ran the same core planner as the webpage: default caps → 61 simulated servings; Tomato & Egg Stir-fry cap 0 → 45. Local verification reproduced both.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides an execution environment for the same core planner used by the webpage and its two reproducible scenario checks. The module uses standard JavaScript and can run in other compatible Node.js environments without additional dependencies. Replit is therefore replaceable as a validation environment.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State “No user data is collected” if applicable.

[Maximum 150 words]

Browser localStorage stores confirmed inventory, reservations, recipes, plan snapshots, activity/import records and optional ingredient icons without automatic expiry. Reset Demo replaces current state; migration backups remain until site data is cleared. Clearing site data removes local copies, not provider records.

Only explicitly submitted extraction sources and task instructions pass through Vercel to Gemini. Raw text/images remain in UI memory, not persistent app storage. The API key is server-only, excluded from frontend builds, and redacted from errors. IP counters are held in function memory; bounded redacted failure diagnostics may enter Vercel logs. Provider retention follows provider policies. Requests set store:false for interaction retrieval; Google's free-tier data-use terms still apply.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini may omit or misread fields. Source comparison, editable preview, schema/field validation and explicit confirmation guard saving. Ambiguous dates and unsupported units remain unresolved. Quotas, provider changes or network failures may interrupt extraction; timeouts, safe errors and manual entry are implemented, without automatic retries or paid fallback.

Browser storage may fail, be cleared or conflict across tabs. Versioned validation, migration backups, whole-state writes and conflict/error messages protect existing state; The public endpoint has input limits, same-origin checks and warm-instance rate/concurrency limits.

7. Compliance & Readiness Checklist

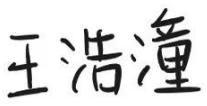
Tick all boxes that apply before submitting.

☑	All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
☑	All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
☑	No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
☑	Any use of generative-AI tools for development has been disclosed and reviewed by the team.
☑	Any use of official SDC sponsor resources has been declared in Section 5.
☑	The prototype described here is accessible to judges through the links provided in Section 1.
☑	All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Leader name: WANG HAOTONG



Electronic signature: _____ _____

Actual signing date: 2026-10-23